AssemblyImpact AI — M04 Slice 1b Implementation, Testing & Evidence Report

Synthetic dataset integration for the EngineeringReview CAP service

Repository	Naman028/assemblyimpact-ai
Branch	feat/m04-slice1b
Implementation commit	68d6501126a1dafcb9f35e0b65a226f11cfc9efb
Base (main)	eaf97f2780d3d66b6bd636aee00d873b29377fdf
Environment	SAP BTP trial — SAP Business Application Studio (dev space ws-lecln)
Pull request	https://github.com/Naman028/assemblyimpact-ai/pull/4
CI	GitHub Actions on 1c53628: success (M03 verification / verify (pull_request): npm ci, lint, test, audit all success, run 42553591994)
Report date	2026-11-09
Prepared by	Claude (Anthropic) for Naman Patel — AI-generated; student review pending

All demonstration records in this report are synthetic (fictional NovaMotion Manufacturing). They are not SAP S/4HANA data. Screenshots are unedited captures of the actual BAS session.

1. Executive summary

Objective. Complete a reproducible synthetic manufacturing dataset and make it available through the existing SAP CAP services (M04 Slice 1b), without starting P3 Project Accelerator generation or P4 UI work.

Status. Implemented and verified in SAP Business Application Studio (BAS). Implementation commit `68d6501126a1` on `feat/m04-slice1b`. GitHub Actions on `1c53628`: success (M03 verification / verify (pull_request): npm ci, lint, test, audit all success, run 42553591994). Draft pull request: https://github.com/Naman028/assemblyimpact-ai/pull/4. Not merged.

- CAP development seed SEED-M04-01 in `test/data`: COOL-100 / EC-000473 (DS-ORACLE-01, plant 1000) and a labelled plant-2000 mirror (EC-200473). 66 synthetic rows in eight CSV files (8 entities), all and traceable to versioned fixtures.
- Generator with deterministic keys and a drift test; the committed CSVs are byte-identical to generator output.
- Seed loads with `npm start` (development); excluded from the test profile by configuration and never resolved in production (verified).
- runImpact on the persisted seed reproduces the independent DS-ORACLE-01 oracle; DS-DEMO-01 is verified through a persisted run in tests only (Decision A).
- Plant isolation, roles, authentication, read-only and malformed-input behaviour re-verified on seeded records.
- BAS: npm ci, lint, 35/35 tests and npm audit (0 vulnerabilities) all exit 0.

Remaining limitations. No Fiori UI; no real S/4HANA data or connection; no Joule Skill or agent; no BTP deployment, HANA persistence or xs-security role definitions; development credentials are public mock values. Synthetic data is not SAP data. Fixture values marked PROPOSED require owner review. The development and test data-loading profiles were checked and the production profile was verified by configuration and test only; **no production deployment has been performed**.

2. Starting state

Item	Value
GitHub baseline	main = eaf97f2780d3 (M04 Slice 1a squash-merge, PR #3); CI passed (owner-reported, PR #3 CI run on ea53c4d verified earlier)
BAS environment	SAP BTP trial, dev space ws-lecln, /home/user/projects/assemblyimpact-ai, synchronized to main (fast-forward)
Runtime (BAS)	Node v24.21.0, npm 11.19.0, @sap/cds 10.2.2, @sap/cds-compiler 7.2.1, @cap-js/sqlite 3.1.2, eslint 9.39.5
Architecture	CAP OData V4: AssemblyCatalog (M03) and EngineeringReview (M04 Slice 1a, read-only + runImpact action, ruleset 1.1.0)
Original problem	Services started empty: the synthetic fixtures were usable by tests only; the development profile had no EngineeringReview users; nothing guarded production against test data.

3. Implementation record

Unit	Files	What	Why / decision	Evidence
U1 Seed builder	test/m04-seed.js	Builds CAP rows from test/fixtures/m04 for each seed scenario; name-based UUID keys; mirror logic with evidence-reference prefix; CSV serialiser.	Single source of truth; reproducibility (D-1b-2, D-1b-5, D-1b-7).	E-02, drift test
U2 Generator CLI	tools/m04/build-seed.mjs, tools/m04/README.md	Writes test/data/*.csv; --check exits 1 when out of date.	Regeneration without hand edits (D-1b-5).	E-02, E-07
U3 Seed CSVs	test/data/assemblyimpact-*.csv (8), test/data/README.md	66 synthetic rows with provenance; README documents origin and loading rules.	Requirement 2, 3, 5, 7 (D-1b-1, D-1b-3, D-1b-9).	E-02, E-03, E-04
U4 Profile configuration	package.json	[test] requires.db.data = db/data, db/csv; development users analyst, cviewer, analyst2000; npm test includes the new suite.	Keep existing suites isolated; demonstrable dev profile (D-1b-4, D-1b-8).	E-02, E-04
U5 Verification suite	test/seed-data.test.js	7 tests: drift, integrity, CAP loading and per-plant OData, oracle via persisted runImpact, auth/plant/input negatives, DS-DEMO-01 persisted run, production exclusion.	Requirements 6-12; quality gates.	E-07
U6 Documentation	readme.md	Slice 1b section, development accounts, evidence statement with AI-assistance disclosure.	Traceability and truthful AI records.	-

Technical explanation. CAP 10.2.2 resolves seed folders from `cds.requires.db.data` (see `lib/dbs/cds-deploy.js` in the installed package). The installed defaults are `db/data, db/csv, test/data` for development and test, and `db/data, db/csv` for production (HANA). The project overrides only the test profile, so the existing M03 and Slice 1a suites keep inserting their own rows into an empty in-memory database, while `test/seed-data.test.js` opts in through `CDS_CONFIG`. Impact runs are never seeded; they are produced by `runImpact`, which needs the seeded `ReviewCases.version` compare-and-set token.

The full decision log (D-1b-1 to D-1b-9) is in `docs/evidence/m04-slice1b/decisions.md`.

4. Dataset specification

Dataset **SEED-M04-01** — fictional company NovaMotion Manufacturing. **Synthetic; not SAP S/4HANA data.** Source: `test/fixtures/m04/DS-ORACLE-01` (v1.0.0, reference date 2026-11-08). Values marked PROPOSED in the data contract still need owner confirmation.

Entity (CSV)	Plant 1000	Plant 2000 mirror	Total	Keys / relationships
Assemblies	1	1	2	COOL-100 per plant; description carries [SYNTHETIC ...]
BOMStates	2	2	4	BOM-COOL-100-A (rev A, 2026-01-01..2026-11-30, RELEASED), -B (rev B, from 2026-12-01, PROPOSED, EC-000473)
BOMItems	7	7	14	A: FAN-A 1, BRKT-01 2, SENS-01 1; B: FAN-B 1, BRKT-01 3, SENS-01 1, SEAL-01 1 (PC)
CostPrices	5	5	10	EUR, price unit 1 PC, SYNTHETIC_STANDARD_PRICE, valid 2026-01-01..9999-12-31
StockSnapshots	5	5	10	snapshot 2026-11-08T08:00:00Z; FAN-A 40, FAN-B 80, BRKT-01 250, SENS-01 120, SEAL-01 150
Instructions	3	3	6	WI-COOL-100-ASM v2 (rev A), v3 (rev B), WI-COOL-100-TST v2 (all revisions)

Entity (CSV)	Plant 1000	Plant 2000 mirror	Total	Keys / relationships
InstructionReferences	9	9	18	step -> material/quantity per instruction version
ReviewCases	1	1	2	EC-000473 / EC-200473: A -> B, production quantity 100, IN_REVIEW
Total	33	33	66	

Prices (EUR/PC): FAN-A 35.00, FAN-B 42.00, BRKT-01 3.00, SENS-01 15.00, SEAL-01 2.00. **Expected result (oracle, ruleset 1.1.0, as of 2026-11-08):** cost per assembly 56.00 -> 68.00 (+12.00), total for 100 assemblies 5600.00 -> 6800.00 (+1200.00, +21.43 %); stock gaps BRKT-01 50 and FAN-B 20; comparison REPLACED / QUANTITY_CHANGED / UNCHANGED / ADDED; one outdated instruction reference (WI-COOL-100-TST step 0030 -> FAN-A); 7 findings (HIGH 3, MEDIUM 2, LOW 2).

Provenance: every row with a sourceType is SYNTHETIC_FIXTURE and carries an evidence reference to its fixture row; mirror references start with `PLANT-2000-MIRROR:`. **Not seeded:** PUMP-200 / DS-DEMO-01 (test-only, Decision A) and impact runs (always computed). **Limitations:** one scenario only; no historical revisions beyond A/B; no S/4HANA identifiers; data contract values marked PROPOSED await owner confirmation.

5. Testing and verification

Command (BAS, E-07)	Exit	Observed
npm ci	0	found 0 vulnerabilities
npm run lint	0	eslint srv test --max-warnings=0
npm test	0	tests 35, pass 35, fail 0, cancelled 0, skipped 0
npm audit --package-lock-only --audit-level=high	0	found 0 vulnerabilities
node tools/m04/build-seed.mjs --check	0	test/data is up to date
git status (after tests)	-	0 tracked changes; no untracked files outside the evidence folder

New tests (test/seed-data.test.js), all passing locally (Node 24.22.2) and in BAS (Node 24.21.0):

Test	Expected	Observed
CSVs equal generator output; generator deterministic	byte-identical	pass
Referential and semantic integrity	unique keys, FKs, plant/assembly agreement, A ends before B, priced and stocked materials, EUR, labels, no PUMP-200	pass
CAP loads test/data; per-plant OData	counts 1/2/7/5/5/3/9 per plant; plant 2000 sees EC-200473 only; M03 shows labelled COOL-100	pass
runImpact on persisted seed vs DS-ORACLE-01	all oracle figures, lines and findings; same fingerprint on repeat	pass
Auth/plant/input negatives on seed	401, 404 cross-plant, 403 viewer/wildcard/malformed/M03 role, 400 bad date, 405 writes	pass
DS-DEMO-01 via persisted runImpact (test-only)	DS-DEMO-01 expected results	pass
Production never loads test/data	production db.data and resources exclude test/data; development includes 8 files	pass

Failures and corrections. (1) Mutation check: changing the seeded FAN-B price from 42.00 to 43.00 made the drift, oracle and price tests fail as intended; restored, all pass. (2) E-06a evidence harness reported 400 for DELETE ImpactRuns because the run ID variable was empty (subshell);

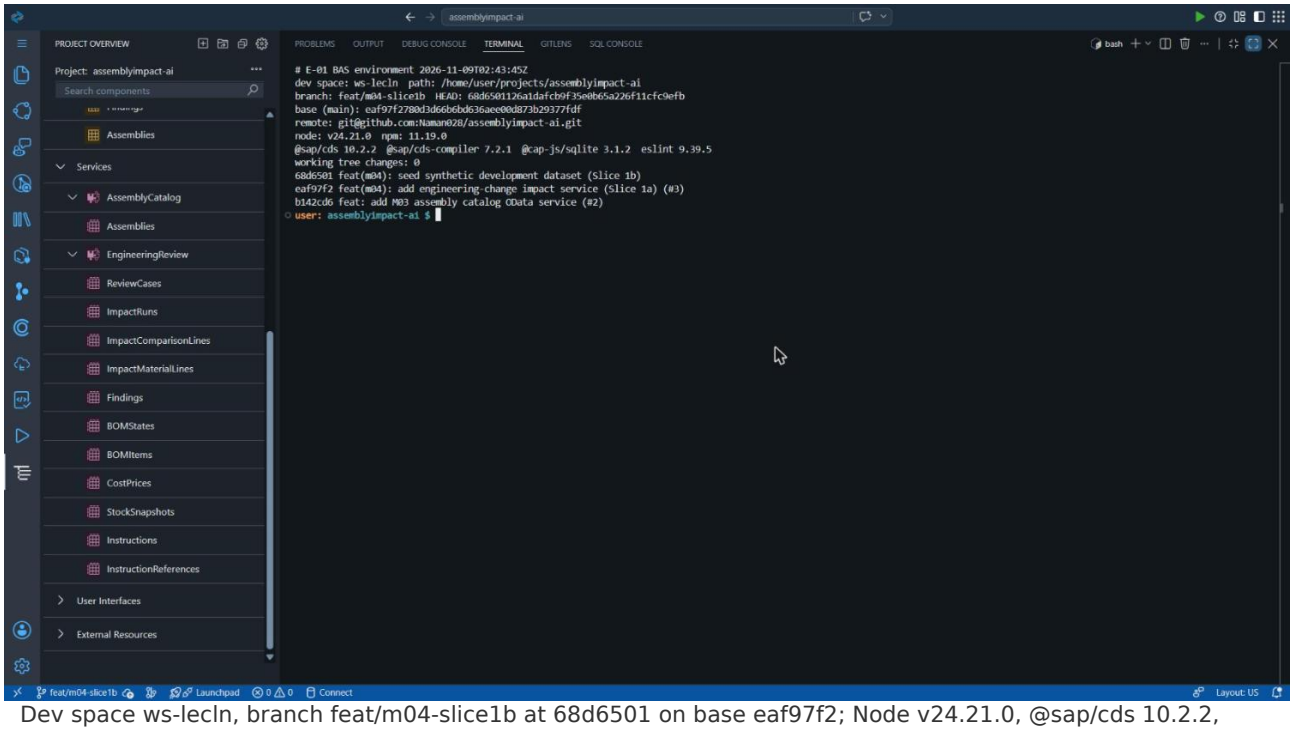
re-run in E-06b returned 405 for DELETE and PATCH. No application code was changed for either.

Live checks in BAS (development profile, E-03 to E-06b): 13 HTTP checks plus the oracle comparison; all pass after the E-06b correction.

6. Screenshot evidence

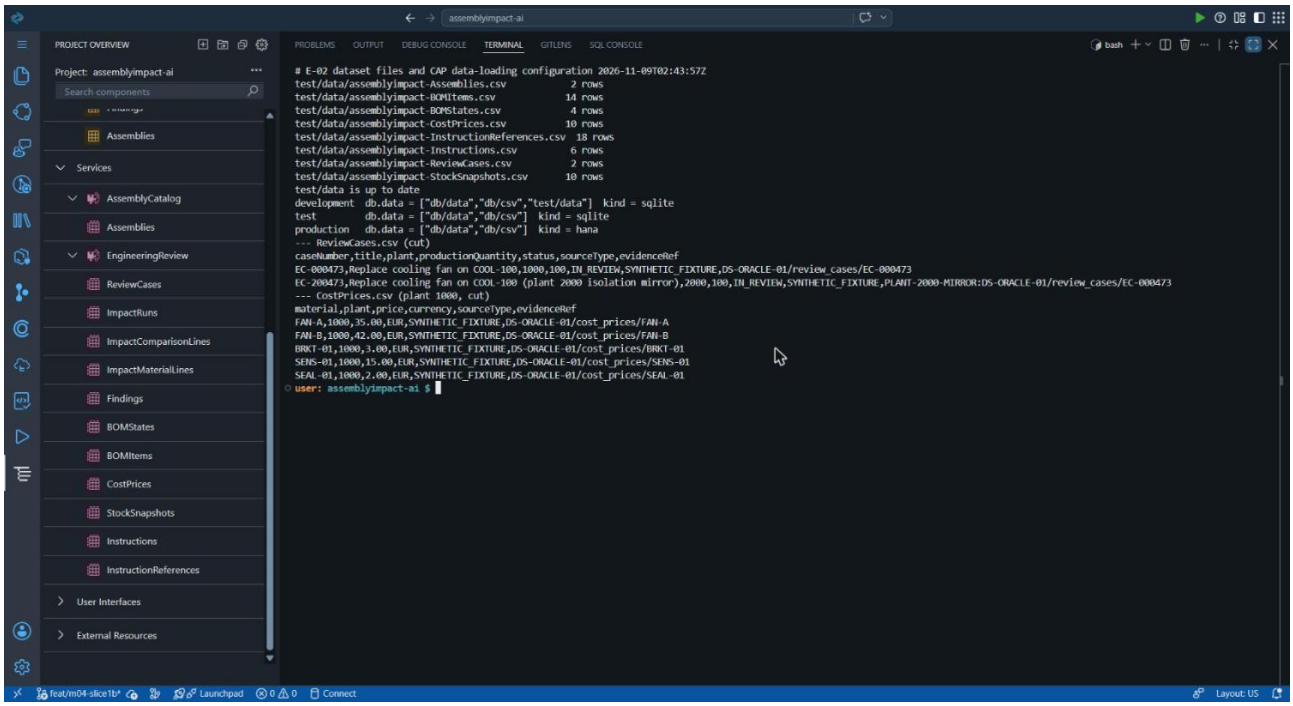
Original, unedited screenshots of the BAS browser tab captured with Claude in Chrome. Times are UTC from the terminal clock. No secrets are visible (no credentials are printed; development passwords are public mock values).

E-01 — BAS repository, branch and environment identification



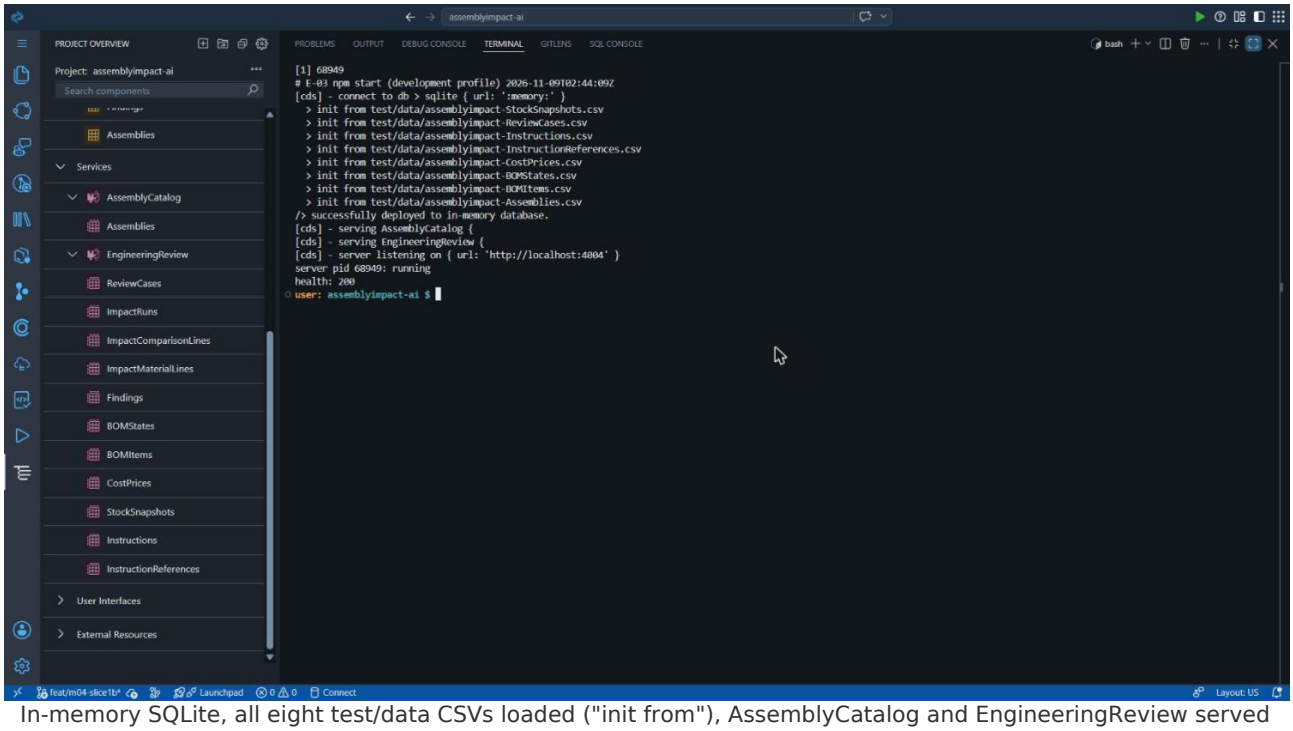
Dev space ws-lecln, branch feat/m04-slice1b at 68d6501 on base eaf97f2; Node v24.21.0, @sap/cds 10.2.2, cds-compiler 7.2.1, @cap-js/sqlite 3.1.2; clean tree.
Captured 2026-11-09T02:43:45Z UTC · file screenshots/E-01_bas-environment.jpg · SHA-256 4e95305d38046deb...

E-02 — CAP development/test dataset files and data-loading configuration



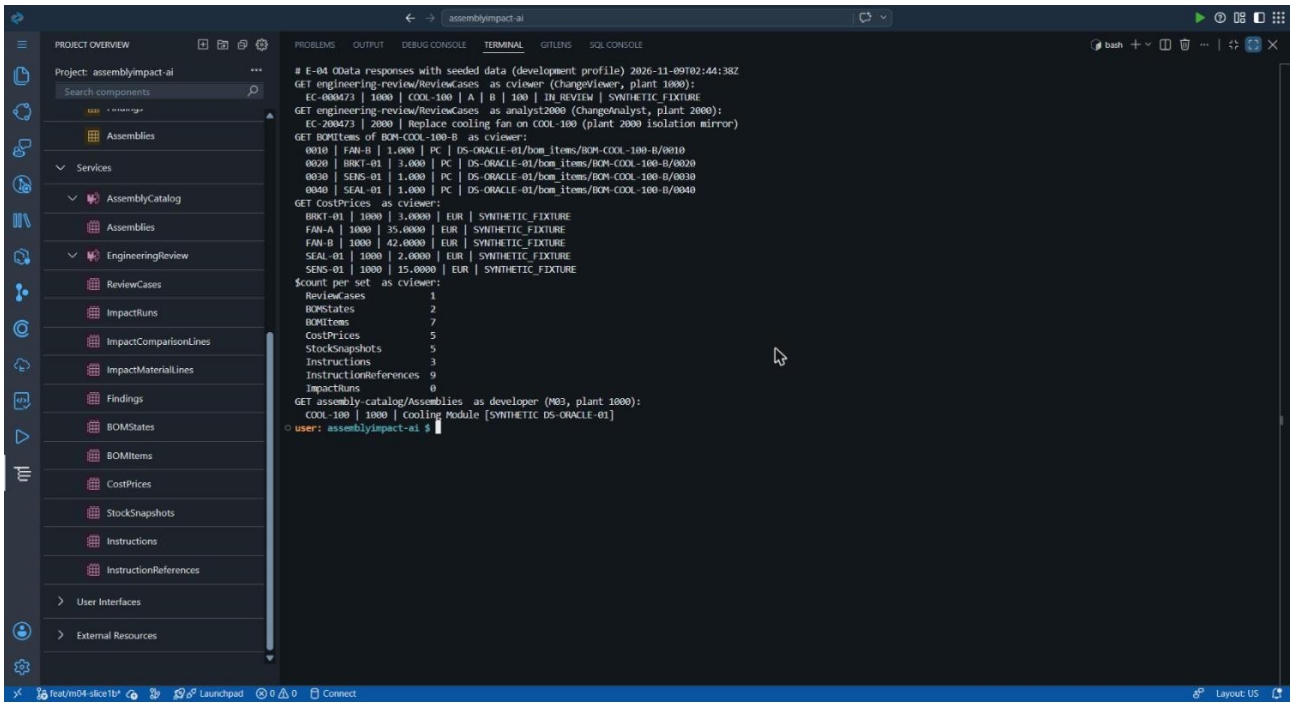
Eight CSVs in test/data with row counts; drift check up to date; resolved db.data: development includes test/data, test and production do not.
Captured 2026-11-09T02:43:57Z UTC · file screenshots/E-02_dataset-files-and-config.jpg · SHA-256 f5cd022e8b1a0e47…

E-03 — Successful CAP server startup (npm start, development profile)



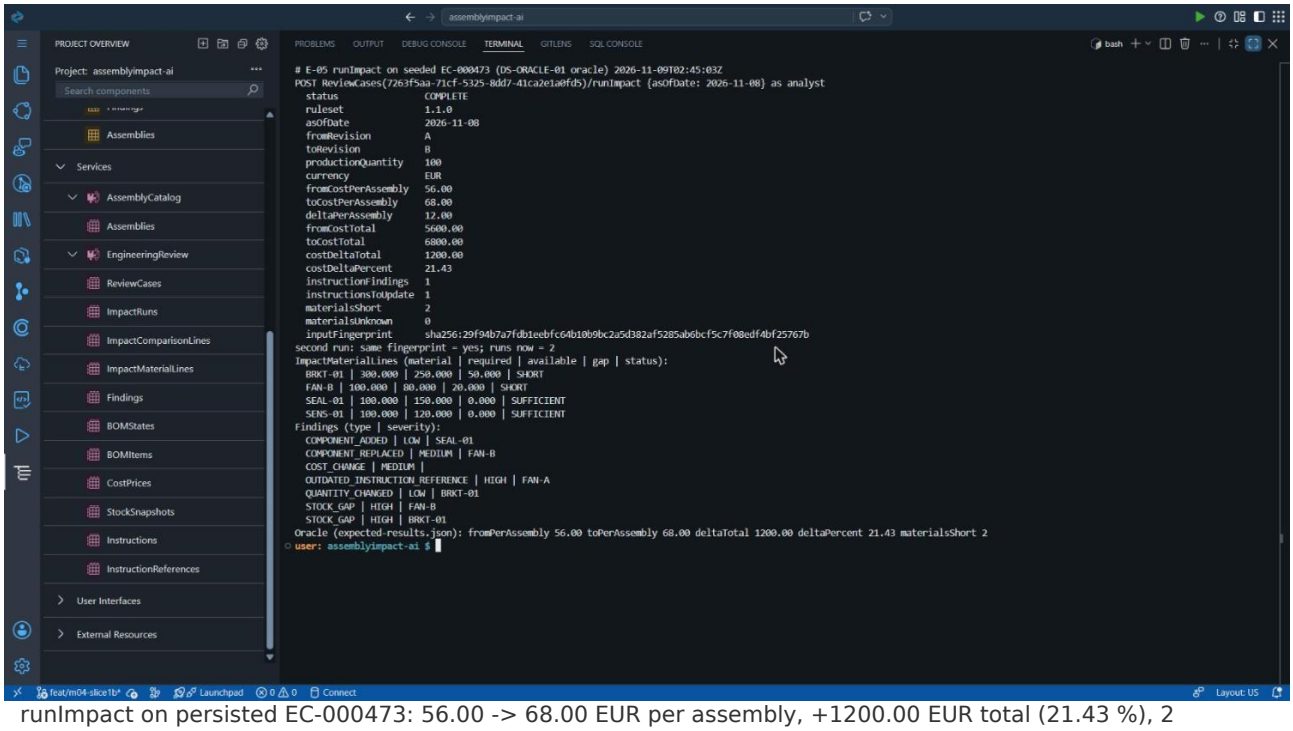
In-memory SQLite, all eight test/data CSVs loaded ("init from"), AssemblyCatalog and EngineeringReview served on :4004, health 200.
Captured 2026-11-09T02:44:09Z UTC · file screenshots/E-03_cap-server-startup-seed-loaded.jpg · SHA-256 b58e58ff6bec997c…

E-04 — OData responses showing seeded engineering data



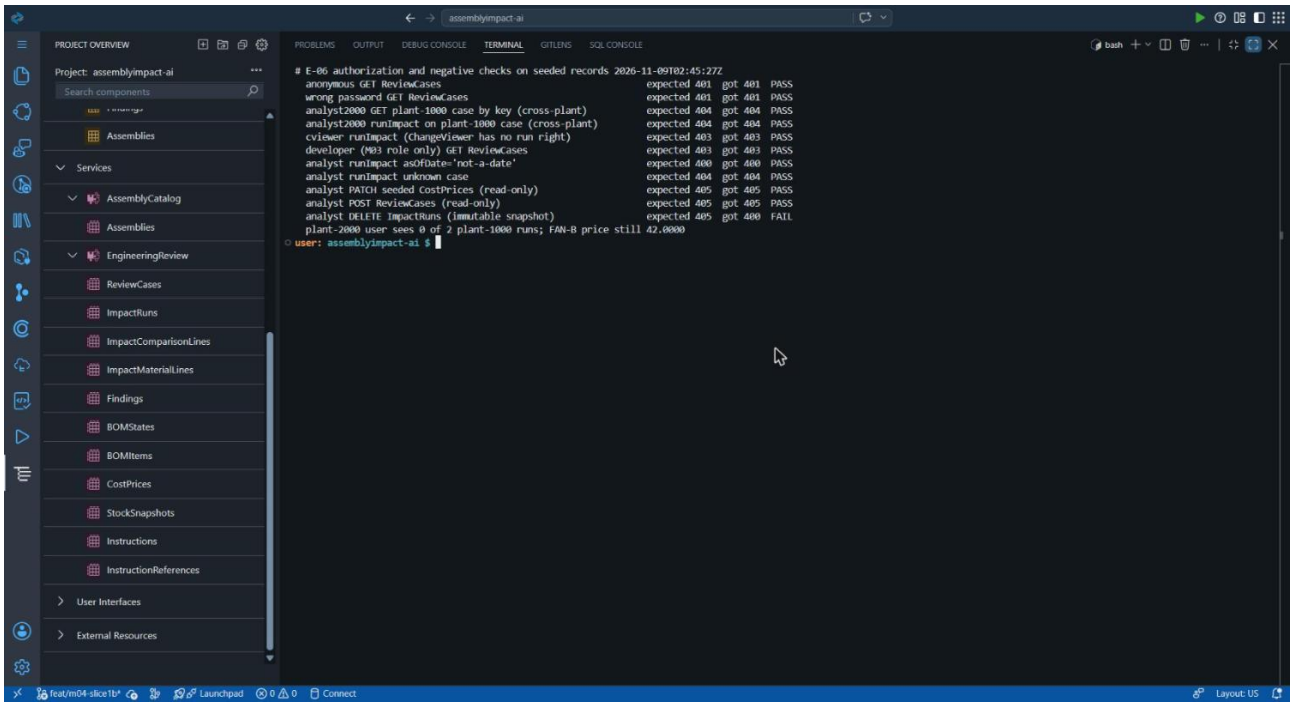
cviewer (plant 1000) sees EC-000473 only; analyst2000 (plant 2000) sees EC-200473 only; BOM B items, prices, $count per set (ImpactRuns 0 before any run); M03 Assemblies shows the labelled COOL-100.
Captured 2026-11-09T02:44:38Z UTC · file screenshots/E-04_odata-seeded-records.jpg · SHA-256 5395d4a6aec34b84…

E-05 — Deterministic impact calculation for the approved oracle scenario



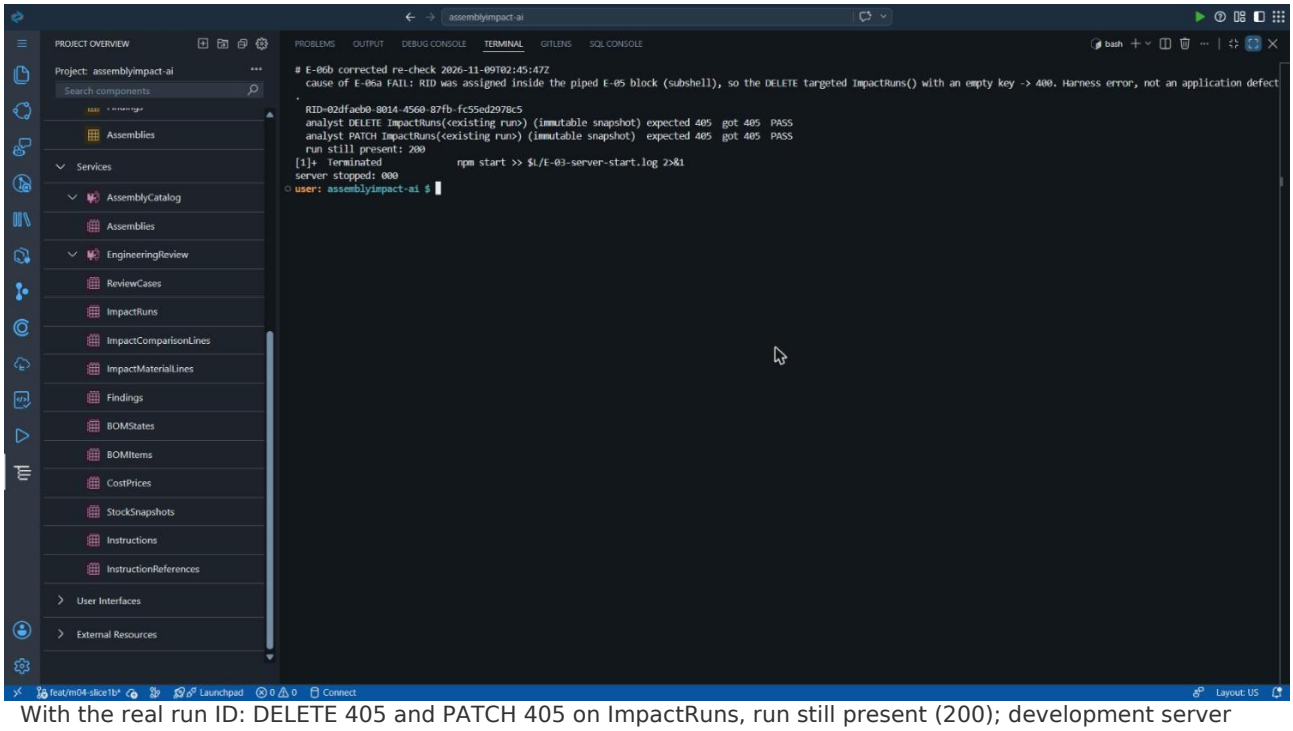
runImpact on persisted EC-000473: 56.00 -> 68.00 EUR per assembly, +1200.00 EUR total (21.43 %), 2 materials short, 7 findings; second run has the same fingerprint.
Captured 2026-11-09T02:45:03Z UTC · file screenshots/E-05_runimpact-oracle-result.jpg · SHA-256 4a05a85d639d1818…

E-06a — Authorization and negative tests (first pass)



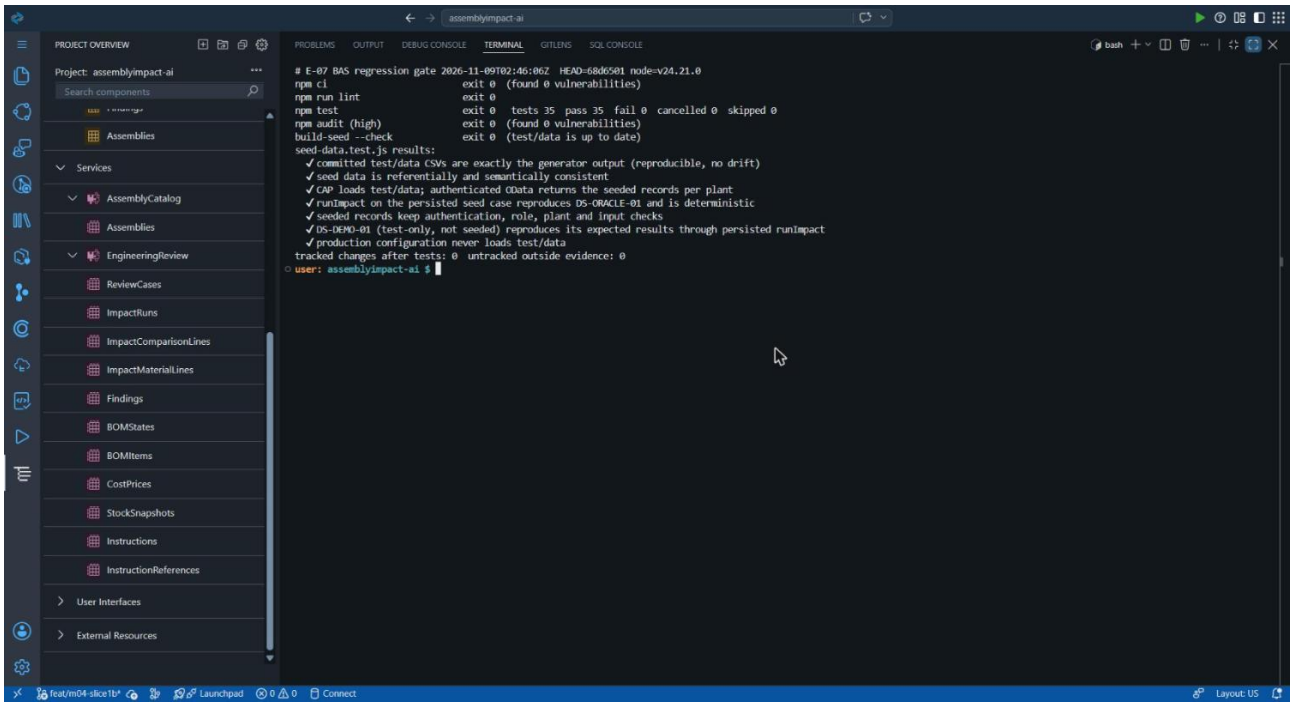
10 of 11 checks PASS. The DELETE ImpactRuns check shows FAIL (400) because of a harness error (empty run ID); kept unedited as evidence.
Captured 2026-11-09T02:45:27Z UTC · file screenshots/E-06a_authorization-negative-checks.jpg · SHA-256 3994af137d1f1276...

E-06b — Corrected re-check of the immutable-snapshot checks



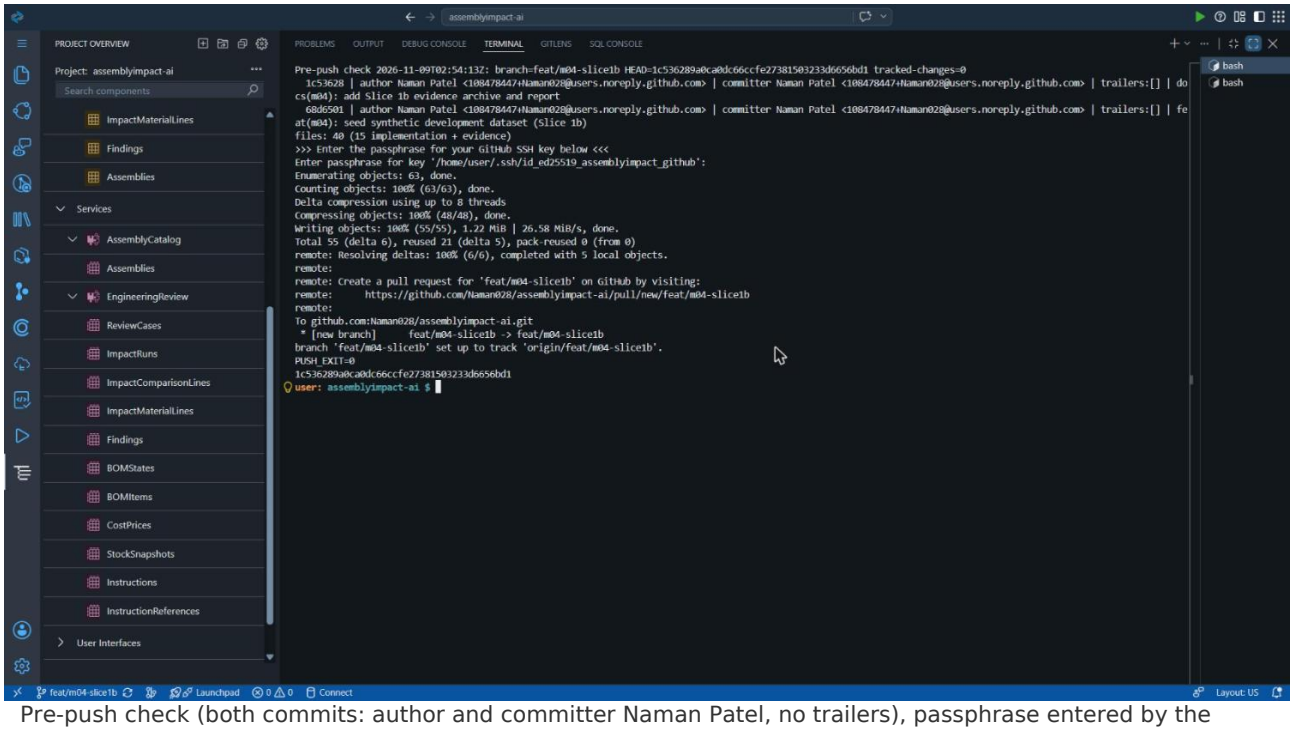
With the real run ID: DELETE 405 and PATCH 405 on ImpactRuns, run still present (200); development server stopped.
Captured 2026-11-09T02:45:47Z UTC · file screenshots/E-06b_authorization-recheck.jpg · SHA-256 77a55457a49d236f...

E-07 — Final BAS regression results



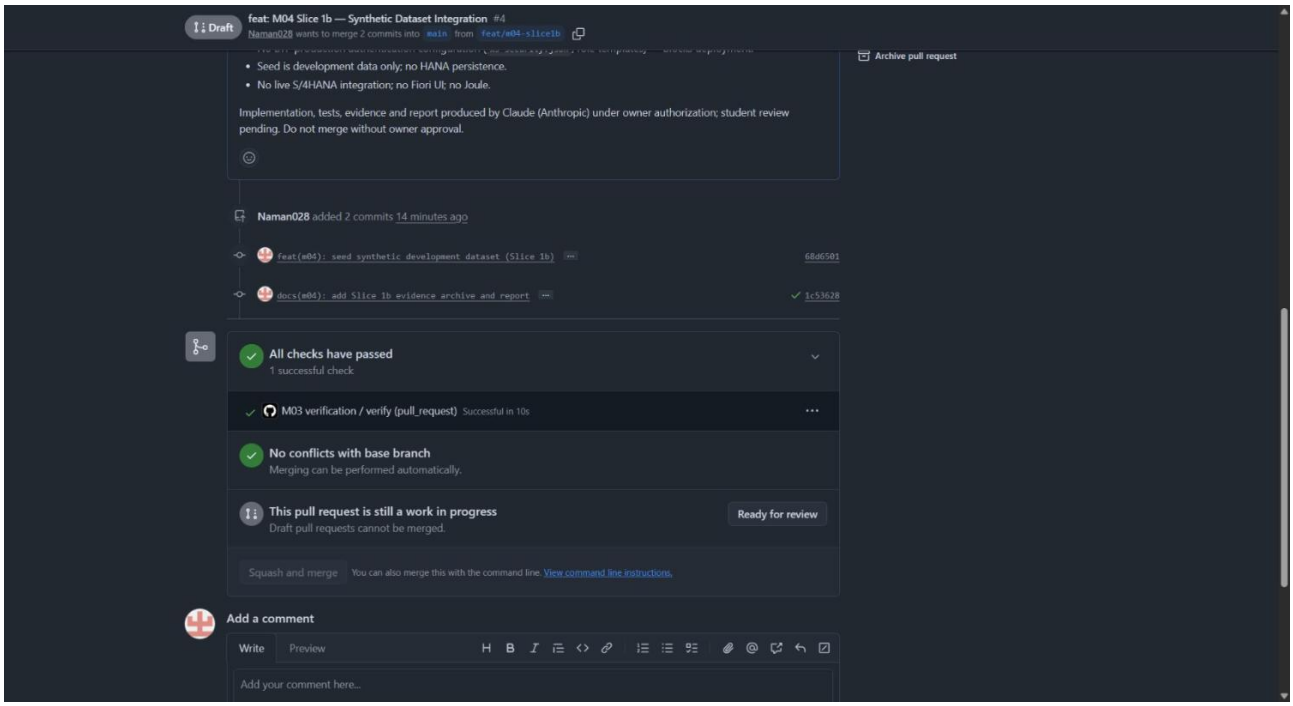
npm ci 0, lint 0, npm test 35/35 (0 fail), npm audit 0 vulnerabilities, build-seed --check up to date, no tracked changes after tests.
Captured 2026-11-09T02:46:06Z UTC · file screenshots/E-07_bas-regression-gate.jpg · SHA-256 82795b6e0126e254...

E-08a — GitHub publication: normal push of the feature branch from BAS



Pre-push check (both commits: author and committer Naman Patel, no trailers), passphrase entered by the owner (not shown), new branch feat/m04-slice1b at 1c53628, PUSH_EXIT=0. No force push.
Captured 2026-11-09T02:54:13Z UTC · file screenshots/E-08a_github-push.jpg · SHA-256 7bef3b8538a323e0...

E-08b — GitHub draft pull request #4 with CI status



Draft PR into main with commits 68d6501 and 1c53628; All checks have passed (M03 verification / verify, pull_request, 10 s); draft pull requests cannot be merged.
Captured 2026-11-09T02:57:00Z UTC · file screenshots/E-08b_github-draft-pr-ci.jpg · SHA-256 bee74735d5c726c6...

7. Security and quality

Area	Result	Evidence
Authentication	Anonymous and wrong password -> 401; production profile still requires JWT (existing test)	E-06a, tests
Role-based authorization	ChangeViewer cannot run (403); M03-only role has no M04 access (403); writes 405	E-06a/b, tests
Plant isolation	Plant-2000 user sees only EC-200473 and 0 plant-1000 runs; cross-plant key and runImpact -> 404	E-04, E-06a, tests
Malformed input	asOfDate "not-a-date" -> 400; unknown case -> 404; malformed/wildcard plant attributes -> 403	E-06a, tests
Synthetic data containment	Production resolves only db/data and db/csv; automated test	E-02, tests
Dependencies	No new packages; package-lock.json unchanged; npm audit 0 vulnerabilities	E-07
Secrets	Diff scan: no keys/tokens/private keys; only public mock passwords in profile config	review

Unresolved risks. No xs-security.json / role templates (blocks BTP deployment, carried over from M03/Slice 1a); development mock passwords are public by design and must never be used in production; GitHub web squash-merges record "GitHub" as committer (owner rule: committer Naman Patel); the GitHub SSH key needs a passphrase or agent for non-interactive fetch/push.

Review disclosure. Code review and quality assessment were performed by one AI system (Claude, Anthropic). No human or SAP reviewer took part.

8. GitHub traceability

Item	Value	Status
Branch	feat/m04-slice1b (from main eaf97f2)	verified
Implementation commit	68d6501126a1dafcb9f35e0b65a226f11cfc9efb — feat(m04): seed synthetic development dataset (Slice 1b); author and committer Naman Patel <108478447+Naman028@users.noreply.github.com>; no co-author trailer	verified (BAS, E-01)
Evidence commit	1c536289a0ca0dc66ccfe27381503233d6656bd1 — docs(m04): add Slice 1b evidence archive and report (25 files under docs/evidence/m04-slice1b; its PDF was generated before CI and showed CI as pending)	verified
Evidence correction commit	docs(m04): finalize Slice 1b evidence with PR #4 and CI results — replaces this PDF and the Markdown report with the final versions, adds E-08a/E-08b, updates github.json, the verification summary and the manifest. A file cannot contain its own commit hash; see PR #4.	see PR #4
Changed files (implementation)	15 files, +518 / -9: package.json, readme.md, test/data/* (9), test/m04-seed.js, test/seed-data.test.js, tools/m04/build-seed.mjs, tools/m04/README.md	verified
Pull request	https://github.com/Naman028/assemblyimpact-ai/pull/4	verified: open, draft, not merged, base main, 40 files (+1693/-9), author Naman028
CI	GitHub Actions on 1c53628: success (M03 verification / verify (pull_request): npm ci, lint, test, audit all success, run 42553591994)	verified
main	unchanged at eaf97f2780d3	verified unchanged after PR creation

9. Four-project relevance

Project	What this milestone contributes	Not claimed
P1 Joule Skill & Agent	A seeded, authorized OData backend a future skill could call (read cases, run impact).	No Joule access or skill exists.
P2 S/4HANA + GenAI	A clearly labelled synthetic baseline to compare against real S/4HANA data later; SourceType S4_IMPORT reserved.	No S/4HANA connection; synthetic data cannot replace it.
P3 AI-assisted Fiori (Project Accelerator)	Real persisted records for the planned Case Worklist to display.	No Project Accelerator generation was run.
P4 Fiori on BTP	Backend data and verified impact engine for the investigation app.	No UI and no BTP deployment.

10. Remaining work

- Owner review of D-1b-1 to D-1b-9 and confirmation of PROPOSED fixture values.
- Merge of the draft PR after review (separate approval).
- Next milestone (proposed): P3 Case Worklist via Project Accelerator in an isolated sandbox. Acceptance criteria: generation in pa-sandbox only; generated files identified; worklist lists the seeded review case(s) from EngineeringReview with plant isolation intact; every manual correction documented; existing 35 tests still pass.
- BTP readiness: xs-security.json with ChangeViewer/ChangeAnalyst role templates and plants attribute, HANA persistence, approuter/IdP, deployment approval.
- P2 blocker: authorized S/4HANA access. P1 blocker: Joule Studio entitlement.
- Tooling: SSH agent or key configuration in BAS for non-interactive Git; ESLint 9 follow-up from M03.

Appendix A — Traceability matrix

Req.	Requirement (milestone brief)	Implementation	Test / evidence
1	Inspect model, services, fixtures, config	Inspection (section 2)	E-01, E-02
2-3	Consistent dataset reusing DS-ORACLE-01 / DS-DEMO-01	U1, U3; DS-DEMO-01 test-only (D-1b-1)	integrity + DS-DEMO-01 tests
4	Preserve DS-NEG-01 coverage	Unchanged calculator tests	E-07 (35/35)
5-6	test/data; verify data loading vs installed CAP	U3, U4 (D-1b-3, D-1b-4)	E-02, production test
7	Mutually consistent keys, dates, units, currency	U1	integrity test
8-9	Load into CAP; OData returns seeded records	U3, U4	E-03, E-04, loading test
10	Deterministic runImpact from persisted data	existing engine on seed	E-05, oracle test
11	M03 works; plant authorization preserved	mirror (D-1b-2)	E-04, E-06a/b, tests
12	Production free of synthetic records	D-1b-3	E-02, production test

Appendix B — Evidence manifest (screenshots)

ID	File	Captured (UTC)	SHA-256
E-01	E-01_bas-environment.jpg	2026-11-09T02:43:45Z	4e95305d38046deb4a18b625d3ec830ad732913c538656c5754ffadeae48fe7f
E-02	E-02_dataset-files-and-config.jpg	2026-11-09T02:43:57Z	f5cd022e8b1a0e47fca03d70e5d9ead799628f647ec57aa47246a56f78a92652
E-03	E-03_cap-server-startup-seed-loaded.jpg	2026-11-09T02:44:09Z	b58e58ff6bec997cd8589e2dcbd054243c12d36732fa0ad791bc37624726d0c5
E-04	E-04_odata-seeded-records.jpg	2026-11-09T02:44:38Z	5395d4a6aec34b8416fd5844b26bcbaaf1062319430b7ecedaee7268d5151b11
E-05	E-05_runimpact-oracle-result.jpg	2026-11-09T02:45:03Z	4a05a85d639d1818c51eb02f998b9dc44a92a673b7330581b830df665e74ef75
E-06a	E-06a_authorization-negative-checks.jpg	2026-11-09T02:45:27Z	3994af137d1f1276607ebb518db88afc1763f4210e1ea331b5e1b6c67332595c
E-06b	E-06b_authorization-recheck.jpg	2026-11-09T02:45:47Z	77a55457a49d236fb58d4da7e40b69316f8d2e51fb79e39581dbe4ef44423ce3
E-07	E-07_bas-regression-gate.jpg	2026-11-09T02:46:06Z	82795b6e0126e254e1d9222502e314c47ed2dc62c8ca72710d6f5549691307a7
E-08a	E-08a_github-push.jpg	2026-11-09T02:54:13Z	7bef3b8538a323e084412a12b3183345c18f2d171dbed1016d4cb10dbc74d9bf
E-08b	E-08b_github-draft-pr-ci.jpg	2026-11-09T02:57:00Z	bee74735d5c726c68726242c799256ccc5d905f8967a202be6d2d8f8882a3b4c

Logs captured in BAS (docs/evidence/m04-slice1b/logs, E-01 to E-07) are hashed in evidence-manifest.json, generated in BAS when the evidence was committed.

Appendix C — Key log excerpt (BAS, E-07)

```
# E-07 BAS regression gate 2026-11-09T02:46:06Z  HEAD=68d6501 node=v24.21.0
npm ci                          exit 0  (found 0 vulnerabilities)
npm run lint                    exit 0
npm test                        exit 0   tests 35  pass 35  fail 0  cancelled 0  skipped 0
npm audit (high)                exit 0  (found 0 vulnerabilities)
build-seed --check              exit 0  (test/data is up to date)
tracked changes after tests: 0  untracked outside evidence: 0
```

Appendix D — References

- SAP CAP (capire): Providing initial data — https://cap.cloud.sap/docs/guides/databases#providing-initial-data
- Installed source used for verification: node_modules/@sap/cds/lib/dbs/cds-deploy.js (deploy.resources), @sap/cds 10.2.2
- Project artifacts: docs/m04/M04_Data_Contract.md, test/fixtures/m04 (DS-ORACLE-01 v1.0.0, DS-DEMO-01 v1.0.1, DS-NEG-01 v1.1.0), M04 v1.2 decisions (Decision A)
- AI assistance: implementation, tests, evidence capture and this report were produced by Claude (Anthropic, model Claude Opus 5.5) under owner authorization; student review pending.